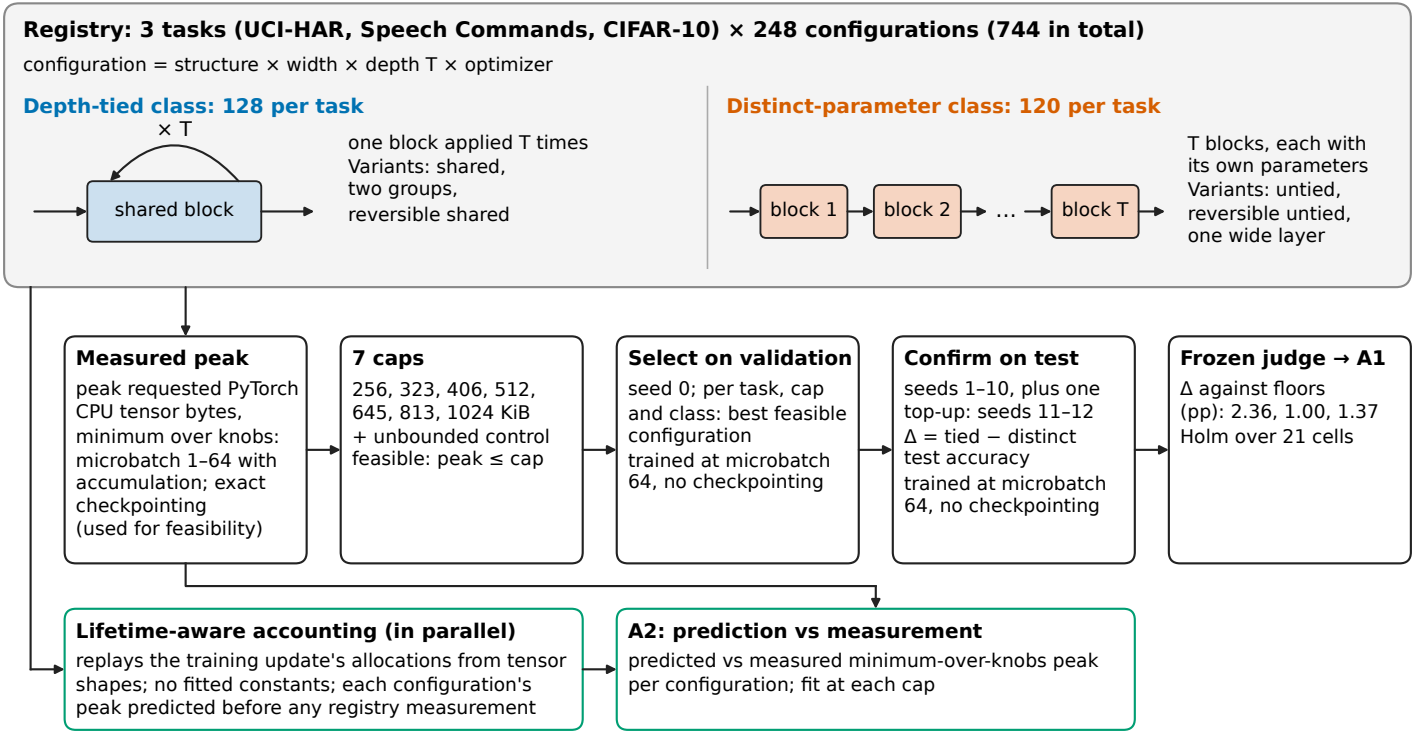
Registry: 3 tasks (UCI-HAR, Speech Commands, CIFAR-10) × 248 configurations (744 in total)
configuration = structure × width × depth T × optimizer
Depth-tied class: 128 per task
× T
shared block
one block applied T times
Variants: shared,
two groups,
reversible shared
Distinct-parameter class: 120 per task
block 1
block 2
…
block T
T blocks, each with
its own parameters
Variants: untied,
reversible untied,
one wide layer
Measured peak
peak requested PyTorch CPU tensor bytes, minimum over knobs: microbatch 1–64 with accumulation; exact checkpointing (used for feasibility)
7 caps
256, 323, 406, 512, 645, 813, 1024 KiB + unbounded control
feasible: peak ≤ cap
Select on validation
seed 0; per task, cap and class: best feasible configuration
trained at microbatch 64, no checkpointing
Confirm on test
seeds 1–10, plus one top-up: seeds 11–12
Δ = tied − distinct test accuracy
trained at microbatch 64, no checkpointing
Frozen judge → A1
Δ against floors (pp): 2.36, 1.00, 1.37
Holm over 21 cells
Lifetime-aware accounting (in parallel)
replays the training update's allocations from tensor shapes; no fitted constants; each configuration's peak predicted before any registry measurement
A2: prediction vs measurement
predicted vs measured minimum-over-knobs peak per configuration; fit at each cap